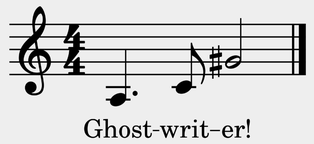
Ghostwriter: Deep Learning and Dynamic Programming for Generating Lyrics from Music

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Problem

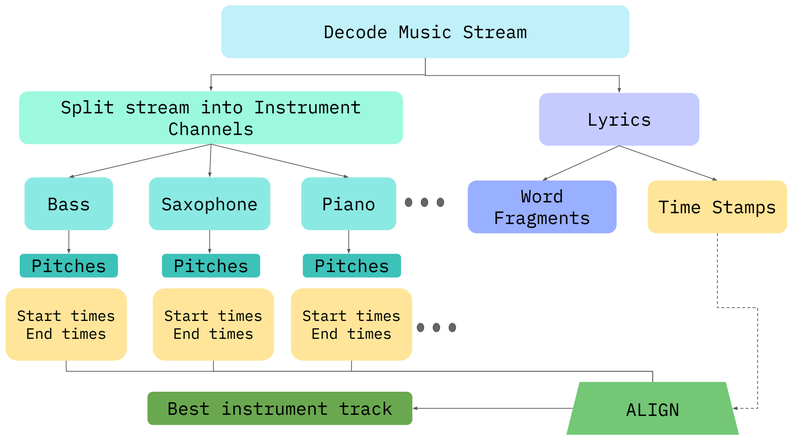
- Many musicians compose music (pitch and rhythm) before writing lyrics (from interviews), but existing music generation models can only generate music when given lyrics or other prompts
- Our goal: generate lyrics that suit a particular input melody (MIDI sequence), matching in the music cadence and mood
- We approach lyric generation as a **Constrained Text Generation** problem (where the syllable count or note timing acts as a constraint) and as a **Sequence-to-sequence Translation problem**.

Background

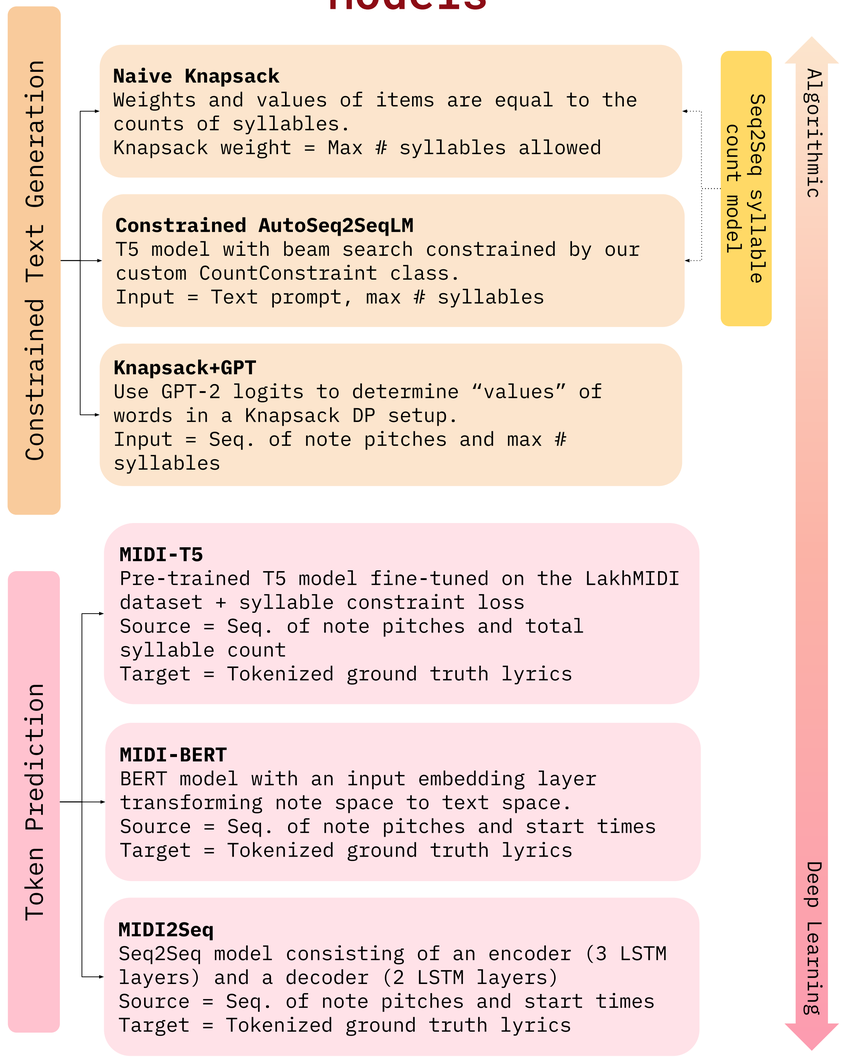
- **MIDI file format** – stores a symbolic representation of music; contains information such as the pitch and duration of notes, contains multiple tracks for various instruments
- **Open-ended text generation** is challenging, lacks standard metrics; interviewing songwriters resulted in three evaluation metrics
- **Existing work** – music and rhythmic prose generation methods:
  - Poetry models leverage sequence models for character level prediction that meets syllabic constraints with theme prompts
  - Instrumental music generation using transformers for note prediction and variational autoencoders to generate latent representations of input music

Dataset

- **LakhMIDI Dataset** – 45,000 MIDI files with timestamped lyrics
- PrettyMIDI used for **note and lyrics extraction** from the MIDI files.
- Since the files did not identify a distinct melody track, we created an **alignment algorithm** to find the instrument track that best aligns with the lyric timing
- We use the best-aligned tracks and their corresponding lyrics as the melody-lyric pairings in the dataset



Models

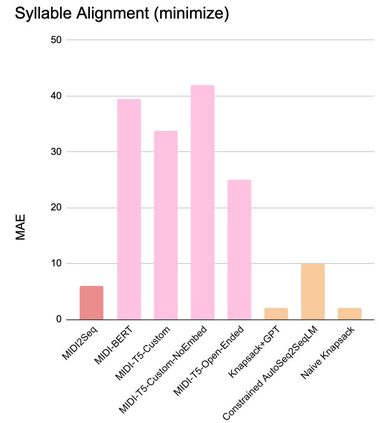


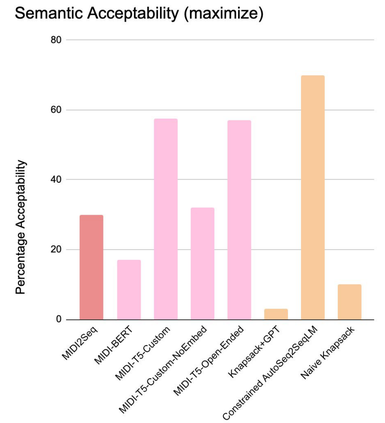
Experiments + Evaluation

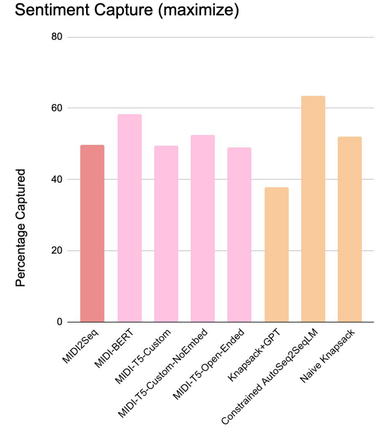
We use three buckets of evaluation to determine lyrical quality (based on interviews conducted with songwriters):

1. **Syllable Alignment** – how well do the syllables of the generated words match the input notes?
2. **Semantic Acceptability** – does the model generate meaningful lyrics with appropriate grammar?
3. **Sentiment Capture** – how well does the sentiment of the generated lyrics align with that of the input notes?

Results







Sample Lyric Generations

MIDI-T5
Goes this way / My heart gets blown away / These eyes are born again / Do you remember how we were? / Do you remem.

Knapsack+GPT
Little point, every point / Mother every other point.

MIDI-BERT
The first time I was a soldier / The last time I was a boy / To live for it / you've got to die for it.

Analysis

- Purely algorithmic models **produce incoherent, grammatically incorrect lyrics**
- Conversely, token prediction models **generate coherent lyrics** with appropriate sentiment, but struggle with syllable alignment
- Strictly-constrained text generation models **perform well in syllable alignment**, but worse in coherence and sentiment
- Constrained AutoSeq2SeqLM – a combination of algorithmic search and deep learning – is the **best performing model across metrics**

Conclusion & Future Work

- Deep learning can **augment algorithmic models** (and vice versa) in constrained text generation settings. As future steps, train on more data and make the Knapsack+GPT model more stable.
- Ethical concerns: filtering text, content attribution, replacing writers